

Oggetto: macchina per innesti su tubi "Varipiper"

Note generali ~~~~~

1_ Specifiche tecniche:

Numero assi interpolati ... 5

Diametro minimo del tubo interno mandrino ... 100 mm

Diametro massimo del tubo interno mandrino .. 610 mm

Diametro minimo del tubo esterno mandrino ... 600 mm

Diametro massimo del tubo esterno mandrino . 1220 mm.

Spessore minimo nominale tubo 4 mm

Spessore massimo nominale tubo .. 50 mm

Lunghezza del tubo lavorabile su due lati interno mandrino 16300 mm

Lunghezza del tubo lavorabile su un lato esterno mandrino 12200 mm

Inclinazione torcia asse X -65/+65 gradi

Inclinazione torcia asse Y -65/+65 gradi

Massimo carico ammesso 12.000 kg

Massimo carico singolo carrello (supporto tubo). 3.000 kg

Tolleranza di lavorazione +1/-1 mm

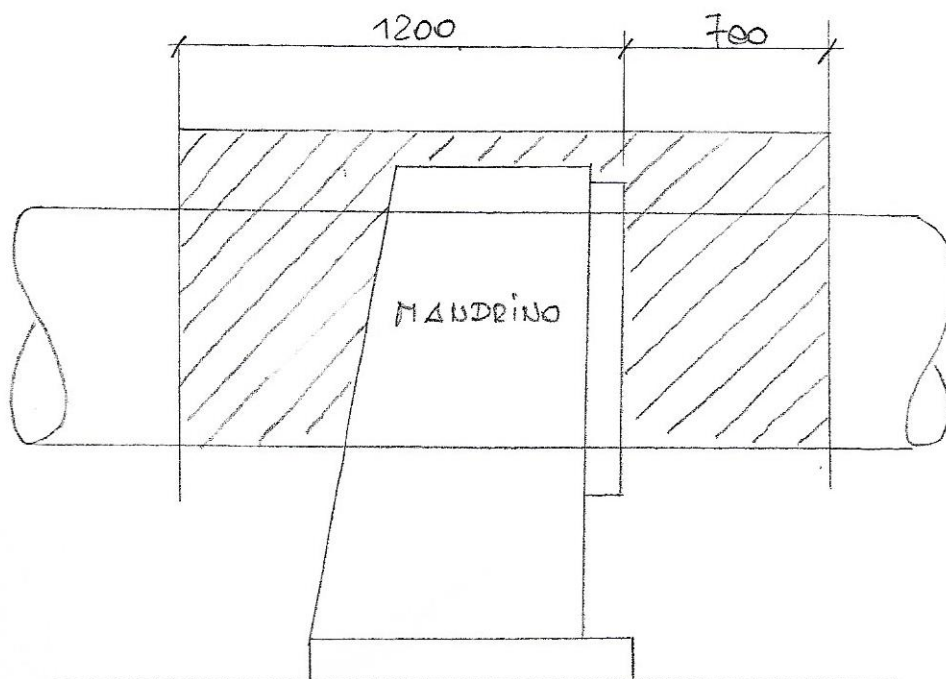
Tolleranza sull'inclinazione degli innesti. +1/-1 grado

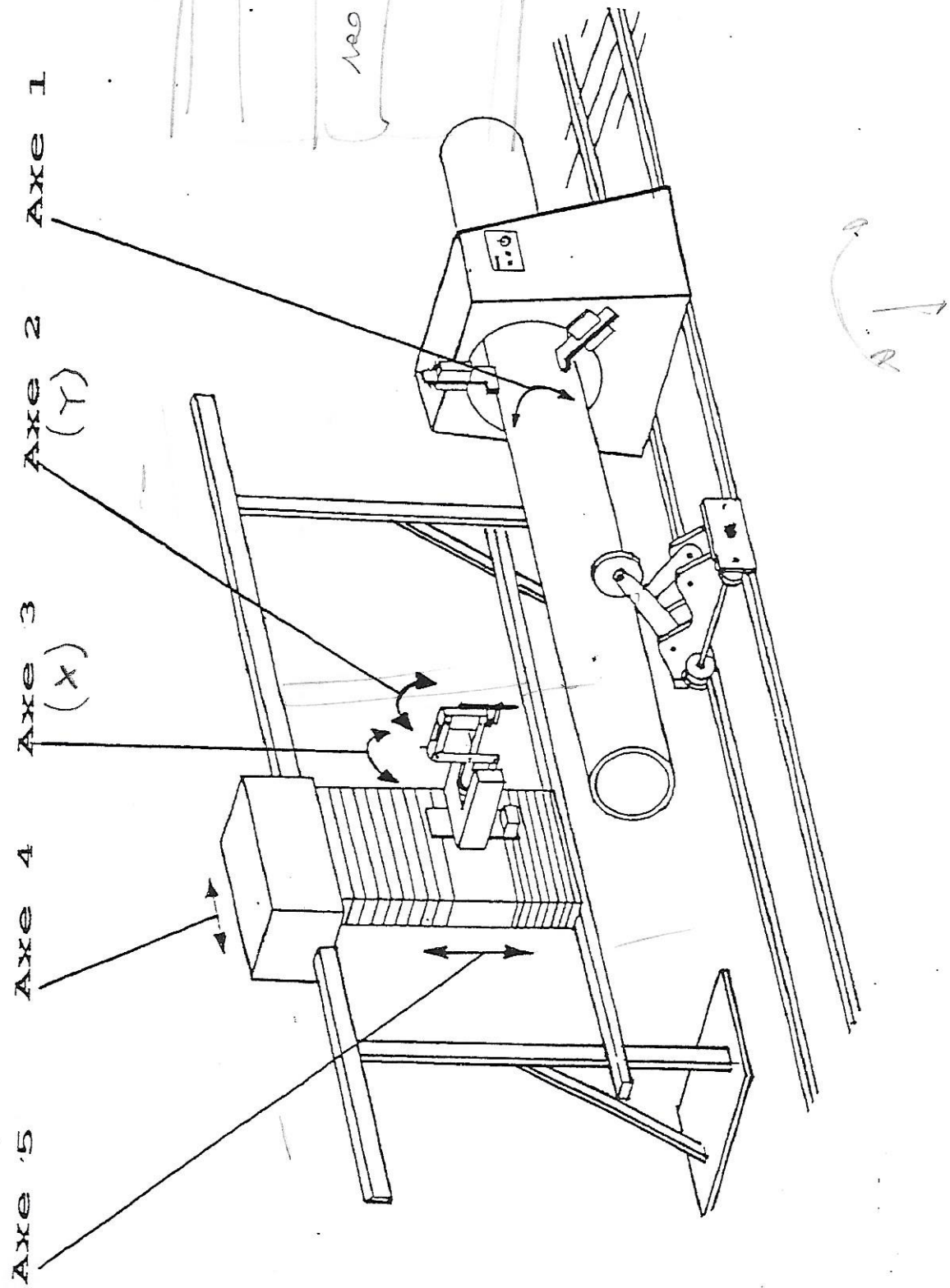
Tolleranza sullo smusso +1/-1 grado

Miscela di taglio Propano/Ossigeno.

Ingombro frontale mandrino .. 700 mm (vedi fig. 1)

Ingombro posteriore mandrino .1200 mm (vedi fig. 1)





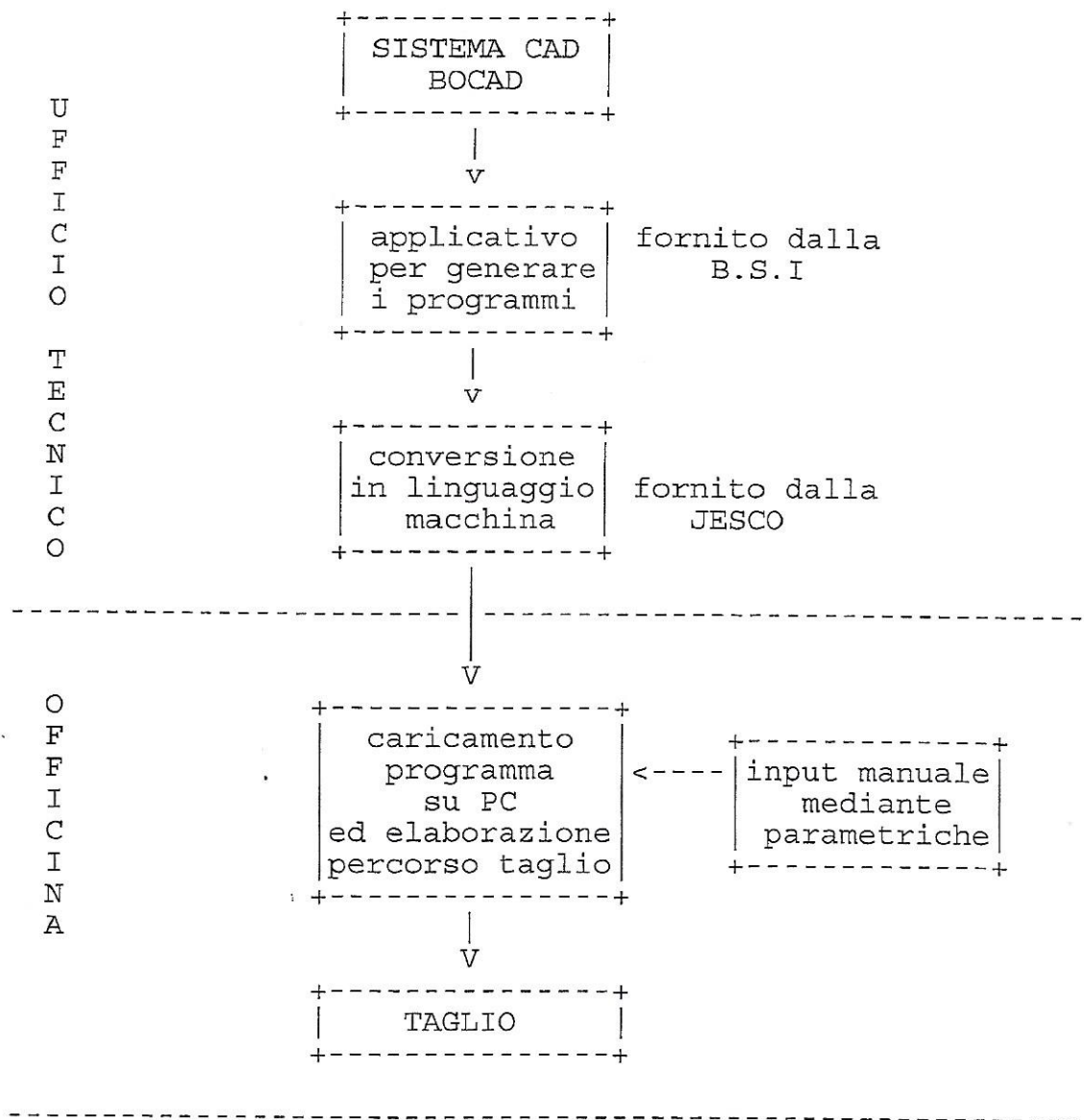
50-0001

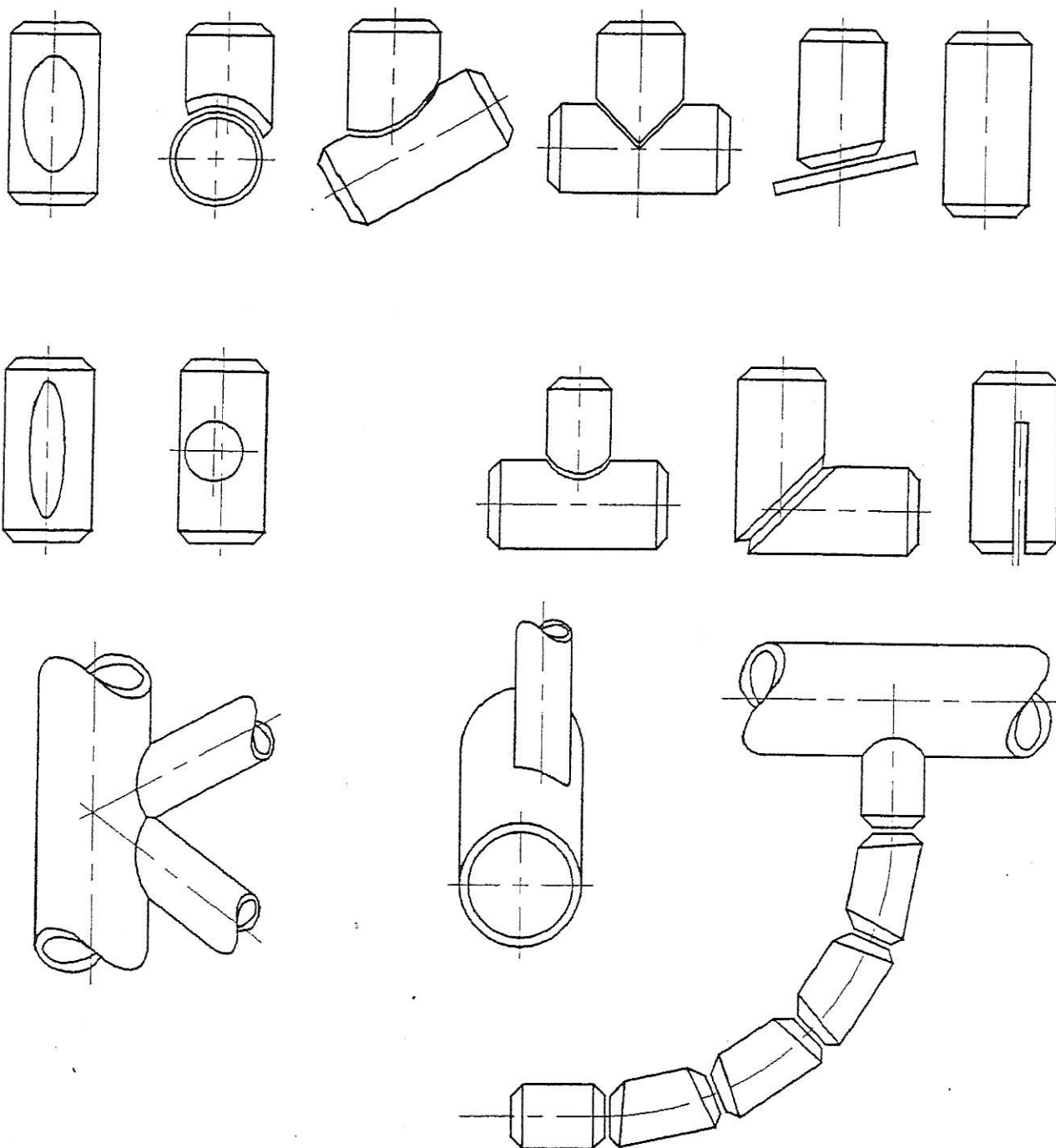
2_ Specifiche controllo numerico (CNC)

Microprocessore 486 DX2 66Mhz.
Hard Disk 210 Mb
Floppy 3.5" 1.44 Mb
Memoria 4 Mb ram
Sistema Operativo ... MSDOS 6.22

3_ Programmazione

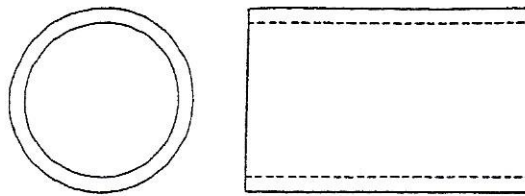
Creazione programmi a bordo macchina mediante PC oppure mediante software B.S.I. su sistema cad BOCAD





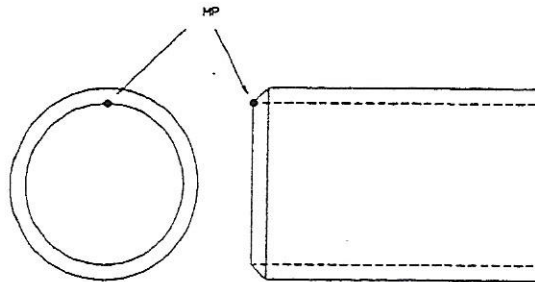
* 01. Straight without bevel.

Parameter: 8. Torch angle
16. Scrapside



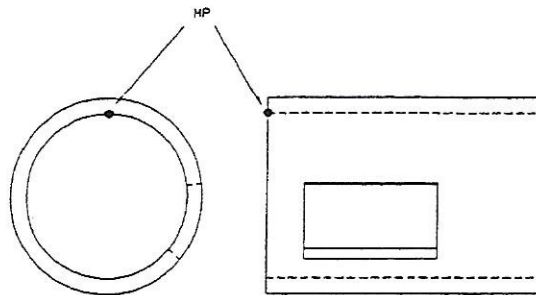
* 02. Straight with bevel.

Parameter: 7. Bevel
8. Torch angle
16. Scrapside
19. Blim. min.
20. Blim. max.



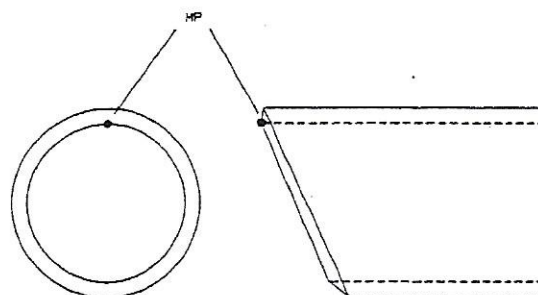
* 03. Slot.

Parameter: 6. Excentricity
7. Bevel
10. Slotlength
11. Slotwidth
16. Scrapside
23. Slot to end.
24. Width angle



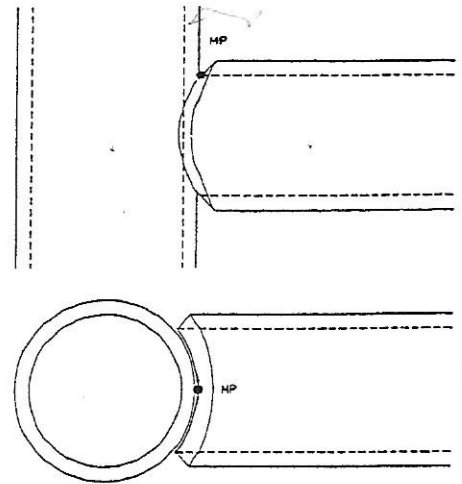
* 04. Mitre.

Parameter: 5. Angle b/m
7. Bevel
8. Torch angle
16. Scrapside
19. Blim. min.
20. Blim. max.



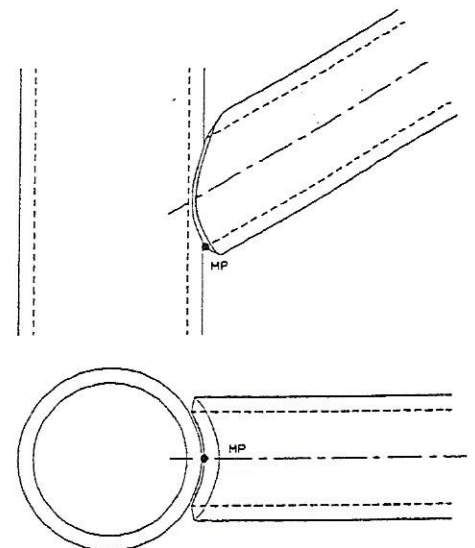
✱ 05. Adaptation, set-on perpendicular on main pipe.

Parameter: 1. Diameter main pipe
 7. Bevel
 8. Torch angle
 16. Scrapside
 19. Blim. min.
 20. Blim. max.



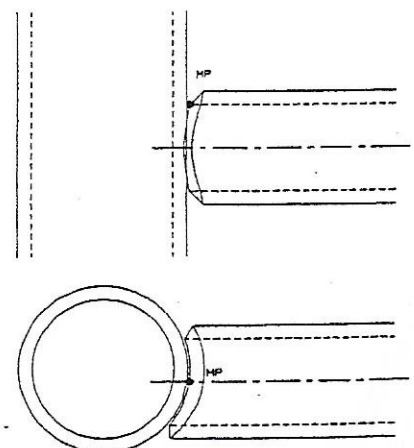
✱ 06. Adaptation, set-on inclined on main pipe.

Parameter: 1. Diameter main pipe
 5. Angle b/m
 7. Bevel
 8. Torch angle
 16. Scrapside
 19. Blim. min.
 20. Blim. max.



✱ 07. Adaptation, set-on excentric perpendicular on main pipe.

Parameter: 1. Diameter main pipe
 6. Excentricity
 7. Bevel
 8. Torch angle
 16. Scrapside
 19. Blim. min.
 20. Blim. max.



* 08. Adaptation, set-on excentric inclined on main pipe.

Parameter: 1. Diameter main pipe

5. Angle b/m

6. Excentricity

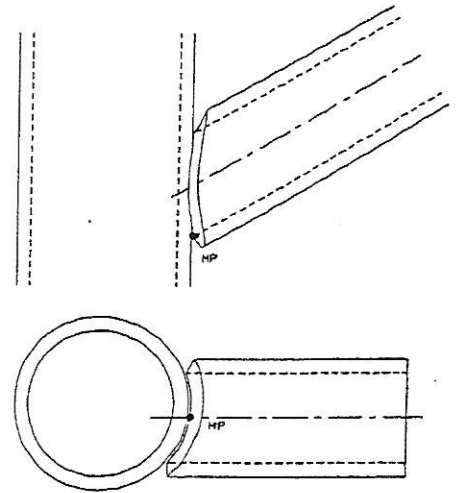
7. Bevel

8. Torch angle

16. Scrapside

19. Blim. min.

20. Blim. max.



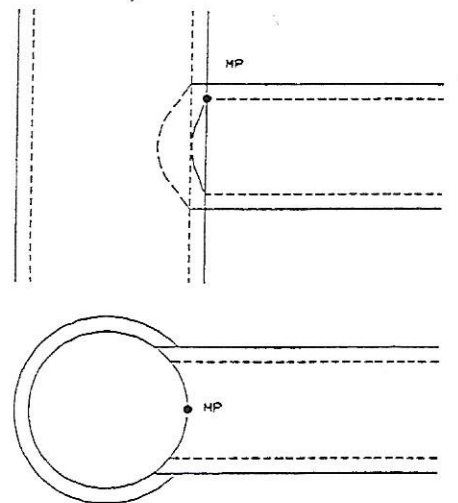
* 09. Adaptation, set-in perpendicular on main pipe.

Parameter: 1. Diameter main pipe

2. Wallthickness main pipe

8. Torch angle

16. Scrapside



* 10. Adaptation, set-in inclined on main pipe.

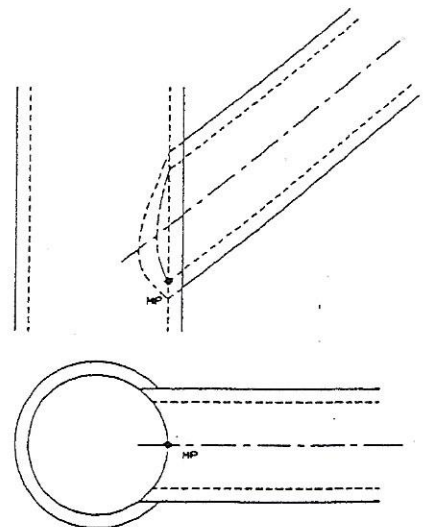
Parameter: 1. Diameter main pipe

2. Wallthickness main pipe

5. Angle b/m

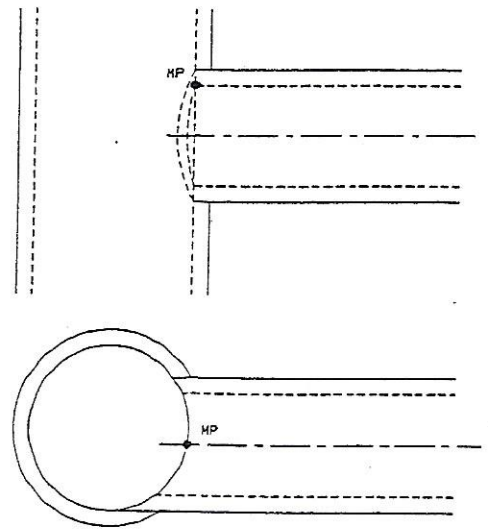
8. Torch angle

16. Scrapside



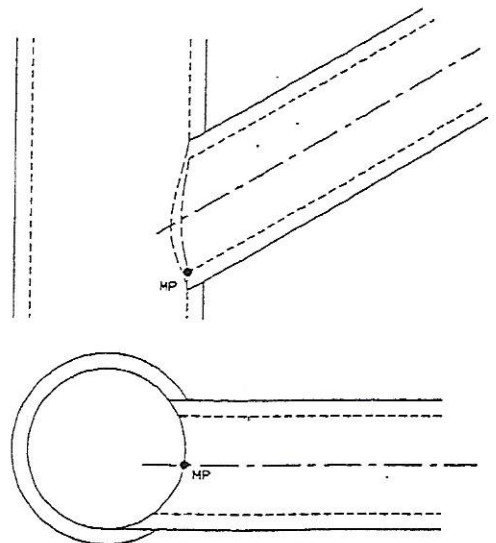
✱ 11. Adaptation, set-in excentric perpendicular on main pipe.

Parameter: 1. Diameter main pipe
 2. Wallthickness main pipe
 6. Excentricity
 8. Torch angle
 16. Scrapside



✱ 12. Adaptation, set-in excentric inclined on main pipe.

Parameter: 1. Diameter main pipe
 2. Wallthickness main pipe
 5. Angle b/m
 6. Excentricity
 8. Torch angle
 16. Scrapside



* 13. Adaptation, offshore inclined on main pipe.

Parameter: 1. Diameter main pipe

5. Angle b/m

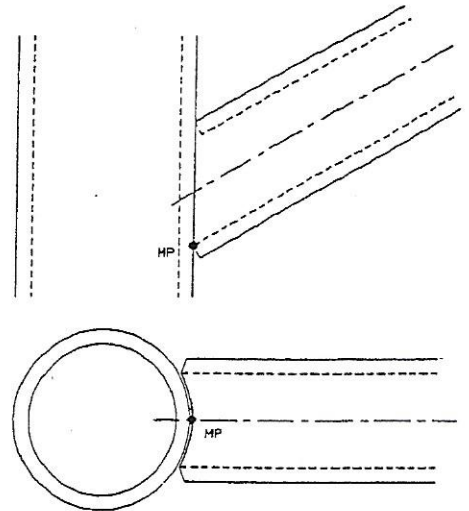
7. Bevel

8. Torch angle

16. Scrapside

19. Blim. min.

20. Blim. max.



* 14. Adaptation, offshore inclined excentric on main pipe.

Parameter: 1. Diameter main pipe

5. Angle b/m

6. Excentricity

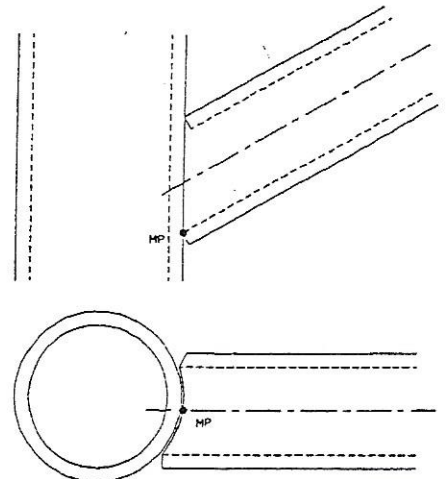
7. Bevel

8. Torch angle

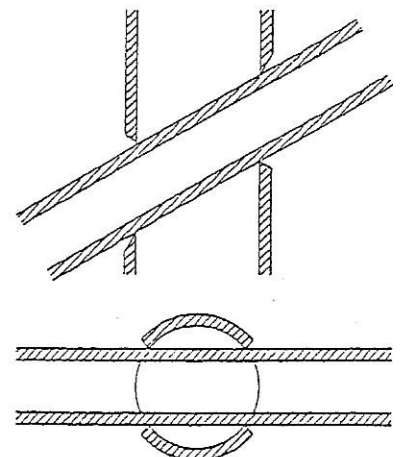
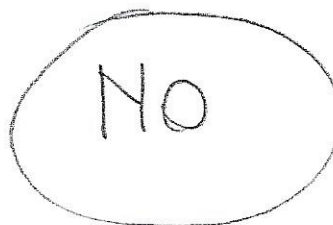
16. Scrapside

19. Blim. min.

20. Blim. max.



15. ISB Rohr durch Rohr.

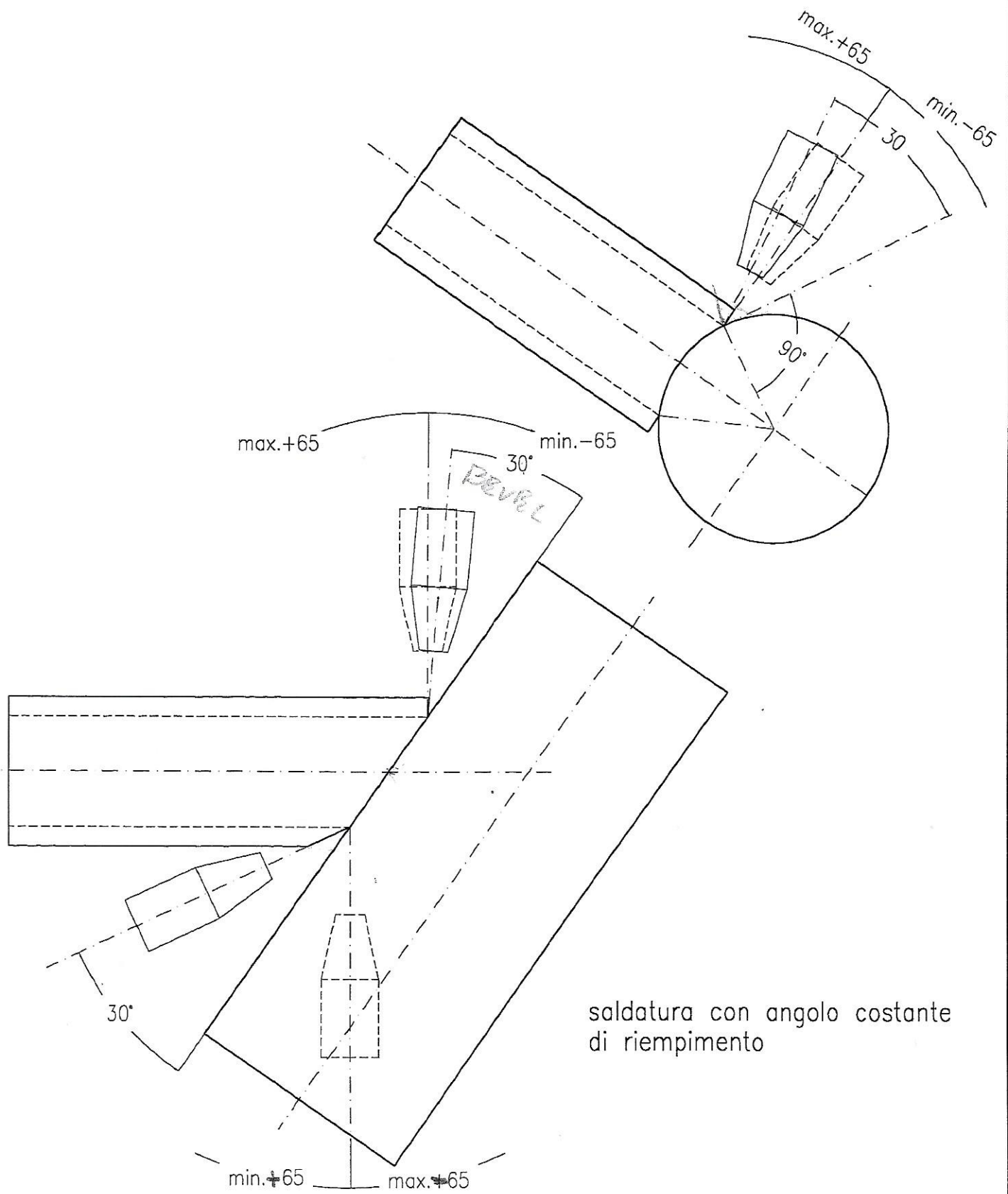


4_ Perparazioni bordi per saldatura

La programmazione permette due tipologie di perparazione:

- a) impostazione di un angolo di riempimento costante di saldatura.
- b) perparazione bordo per saldature denominate "AWS" standard americano (vedi specifiche allegate).

Entrambe le tipologie devono rintrare su innesti che non superino le inclinazioni massime della torcia (-65/+65 gradi per l'asse X e -65/+65 gradi per l'asse Y)



MORE DETAILS ABOUT THE CUTTING SHAPES 13 AND 14 AWS D1.1-92 SECTION 10.13

10.13 Details for Welded Tubular Joints in T-, Y-, and K-Connections Made from One Side Without Backing

These details are subject to the limitation of 10.12.1.

10.13.1 Complete Joint Penetration Groove Welds. Details for complete joint penetration groove welds in tubular T-, Y-, and K-connections used in circular tubes are described in this section. The applicable circumferential range of Details A, B, C, and D are shown in Figure 10.11, and the ranges of local dihedral angles, Ψ , corresponding to them are specified in Table 10.6.

Joint dimensions including groove angles are specified in Table 10.7 and Figure 10.12. When selecting a profile (compatible with fatigue category used in design) as a function of thickness, the guidelines of 10.7.6 shall be observed.

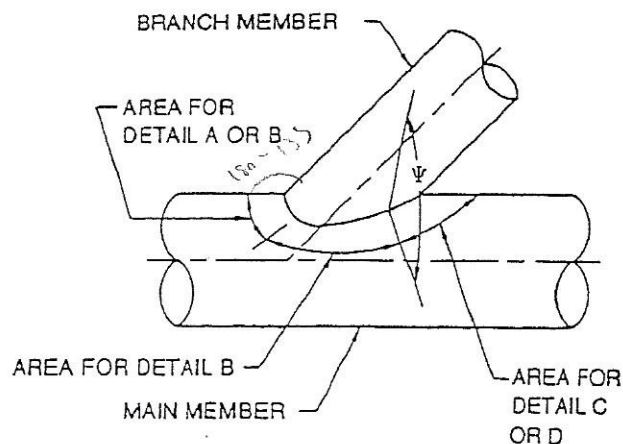


Figure 10.11 — Definitions and Detailed Selection for Complete Joint Penetration Prequalified Tubular Joints for Simple T-, Y-, or K-Connections (see 10.13.1)

Table 10.6
Joint Detail Application (see 10.13.1)

Detail	Applicable Range of Local Dihedral Angle, Ψ
A	180° to 135°
B	150° to 50°
C	75° to 30°
D	40° to 15°

Not prequalified for
groove angles under
30°

Notes:

- The applicable joint detail (A, B, C, or D) for a particular part of the connection is determined by the local dihedral angle, Ψ , which changes continuously in progressing around the branch member.
- The angle and dimensional ranges given in Detail A, B, C, or D include maximum allowable tolerances.
- See Appendix B for definition of local dihedral angle.

Alternative weld profiles that may be required for thicker sections are shown in Figure 10.13. In the absence of special fatigue requirements, these profiles are applicable to branch thicknesses exceeding 5/8 in. (15.9 mm).

Improved weld profiles meeting the requirements of 10.7.5 and 10.7.6 are shown in Figure 10.14. In the absence of special fatigue requirements, these profiles are applicable to branch thicknesses exceeding 1-1/2 in. (38.1 mm). (Not required for static compression loading).

Prequalified details for complete joint penetration groove welds in tubular T-, Y-, and K-connections, utilizing square and rectangular (box) sections, are further described in Figure 10.15. The foregoing details are subject to the limitation of 10.12.2.1.

MORE DETAILS ABOUT THE CUTTING SHAPES 13 AND 14 AWS D1.1-92 SECTION 10.13

Table 10.7

Prequalified Joint Dimensions and Groove Angles for Complete Joint Penetration Groove Welds in Tubular T-, Y-, K-Connections Made by Shielded Metal Arc, Gas Metal Arc (Short Circuiting Transfer)³ and Flux Cored Arc Welding (see 10.13.1)

	Detail A $\Psi = 180^\circ - 135^\circ$		Detail B $\Psi = 150^\circ - 50^\circ$		Detail C $\Psi = 75^\circ - 30^\circ^{***}$	Detail D $\Psi = 40^\circ - 15^\circ^{***}$
End preparation (ω)			90°*		*	
			10° or 45° for $\Psi > 105^\circ$		10°	
Fitup or root opening (R)	FCAW-SS SMAW (1)	GMAW-S FCAW-G (2)	FCAW-SS SMAW (1)	GMAW-S FCAW-G (2)	*** W max. ϕ	
				1/4 in. (6 mm) for $\phi > 45^\circ$	FCAW-SS SMAW (1)	$\left\{ \begin{array}{l} 1/8 \text{ in. (3 mm)} \\ 3/16 \text{ in. (5 mm)} \end{array} \right. \begin{array}{l} 25^\circ - 40^\circ \\ 15^\circ - 25^\circ \end{array}$
	max	3/16 in. (5 mm)	3/16 in. (5 mm)	1/4 in. (6 mm)	5/16 in. (8 mm) for $\phi \leq 45^\circ$	
	min	1/16 in. (1.6 mm) No min for $\phi > 90^\circ$	1/16 in. (1.6 mm) No min for $\phi > 120^\circ$	1/16 in. (1.6 mm)	1/16 in. (1.6 mm)	GMAW-S FCAW-G (2)
Joint included angle ϕ			60° for $\Psi \leq 105^\circ$		40°; if more use Detail B	
			37-1/2°; if less use Detail C		1/2 Ψ	
Completed weld	$\geq t_b$		$\geq t_b$ for $\Psi > 90^\circ$ $\geq t_b / \sin \Psi$ for $\Psi < 90^\circ$		$\geq t_b / \sin \Psi$ but need not exceed 1.75 t_b	$\geq 2t_b$
	t_w					
	L	$\geq t_b / \sin \Psi$ but need not exceed 1.75 t_b			Weld may be built up to meet this	

*Otherwise as needed to obtain required ϕ

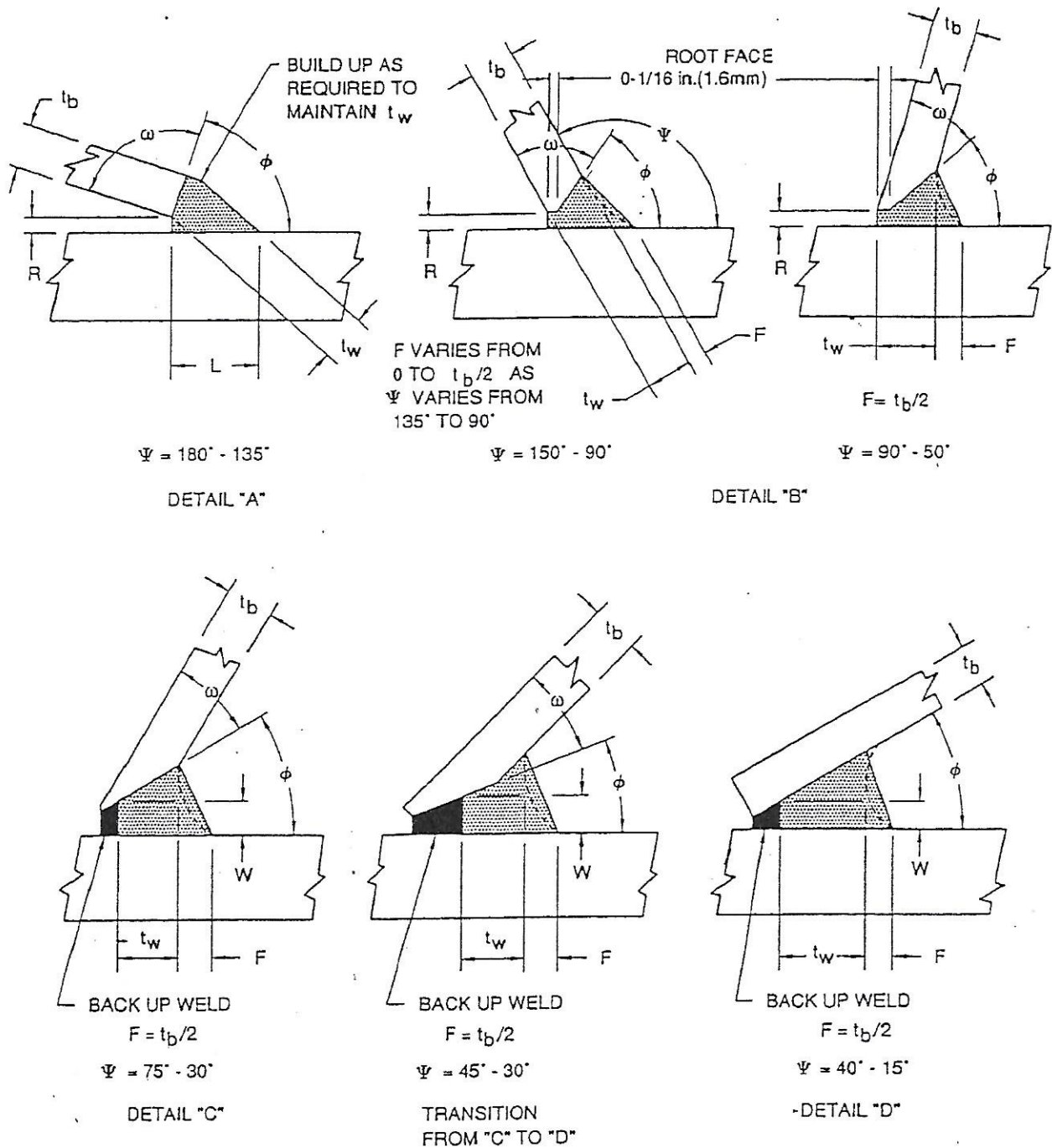
**Not prequalified for groove angles (ϕ) under 30°

***Initial passes of back up weld discounted until width of groove (W) is sufficient to assure sound welding; the necessary width of weld groove (W) provided by back up weld

Notes:

- These root details apply to SMAW and FCAW-SS (self-shielded) in accordance with Table 10.5.
- These root details apply to GMAW-S (short circuiting transfer) and FCAW-G (gas shielded) in accordance with Table 10.5.
- For GMAW-S see 10.12.3.4. These details are not intended for GMAW (spray transfer).
- See Figure 10.12 for minimum standard profile (limited thickness)
- See Figure 10.13 for alternate toe-fillet profile
- See Figure 10.14 for improved profile (see 10.7.5 and 10.7.6)

MORE DETAILS ABOUT THE CUTTING SHAPES 13 AND 14 AWS D1.1-92 SECTION 10.13



Notes:

1. See Table 10.7 for dimensions t_w , L , R , W , ω , ϕ .
2. Minimum standard flat weld profile as shown by solid line.
3. A concave profile, as shown by dashed lines, is also applicable.
4. Convexity, overlap, etc. are subject to the limitations of 3.6.
5. Branch member thickness, t_b , is subject to limitations of 10.7.6.

Figure 10.12 — Prequalified Joint Details for Complete Joint Penetration Groove Welds in Tubular T-, Y-, and K-Connections — Standard Flat Profiles for Limited Thickness (see 10.13.1)

MORE DETAILS ABOUT THE CUTTING SHAPES 13 AND 14 AWS D1.1-92 SECTION 10.13

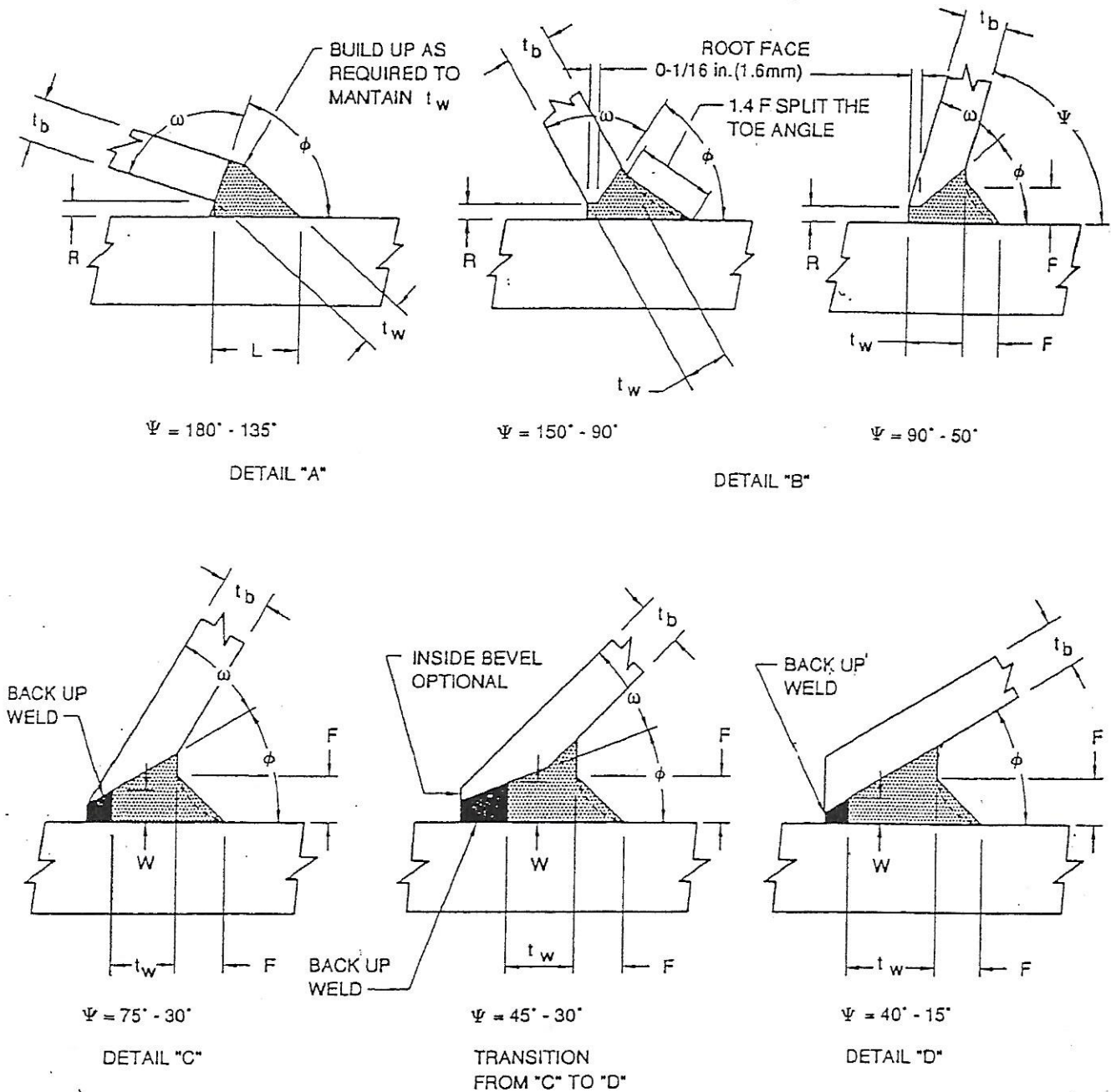
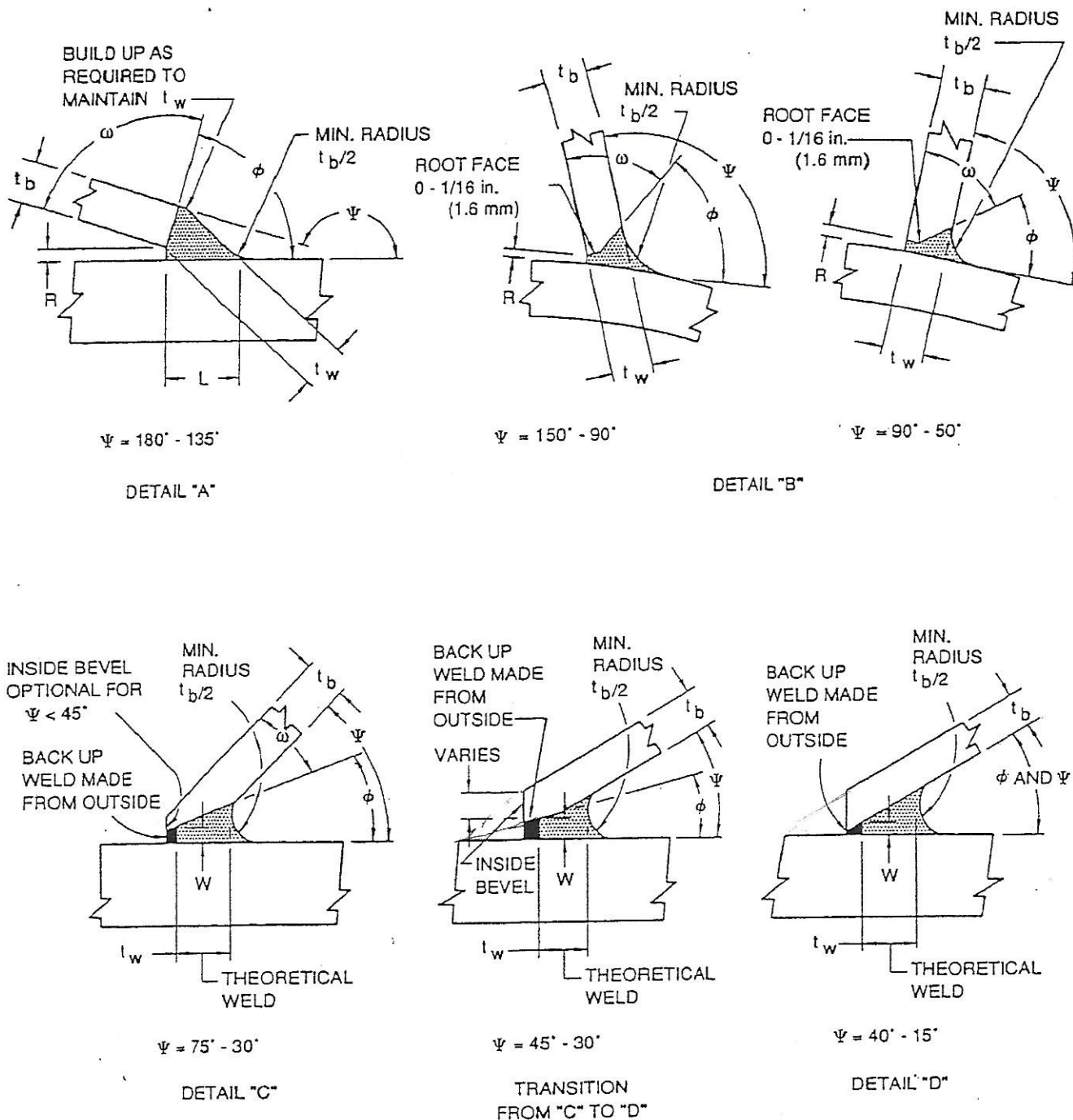


Figure 10.13 — Prequalified Joint Details for Complete Joint Penetration Groove Weld in Tubular T-, Y-, and K-Connections — Profile with Toe Fillet for Intermediate Thickness (see 10.13.1)

MORE DETAILS ABOUT THE CUTTING SHAPES 13 AND 14 AWS D1.1-92 SECTION 10.13



Notes:

1. Illustrating improved weld profiles for 10.7.5(1) as welded and 10.7.5(2) fully ground.
2. For heavy sections or fatigue critical applications as indicated in 10.7.6.
3. See Table 10.7 for dimensions t_b , L , R , W , ω , ϕ .

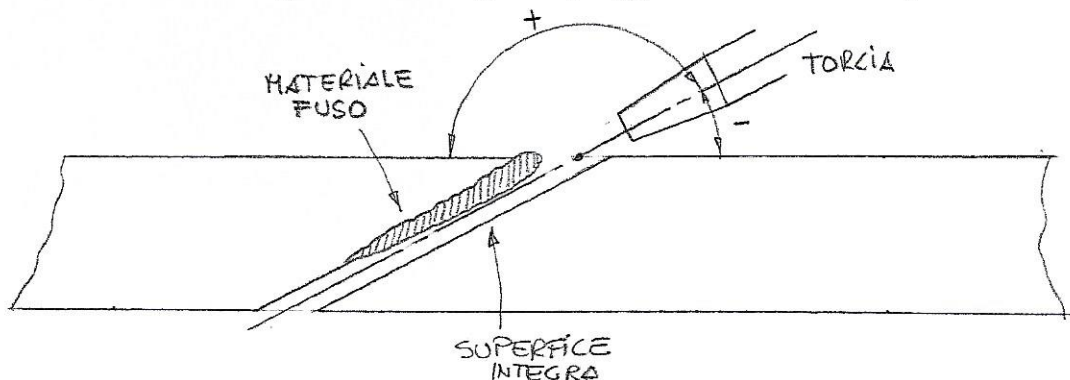
Figure 10.14 — Prequalified Joint Details for Complete Joint Penetration Groove Welds in Tubular T-, Y-, and K-Connections — Concave Improved Profile for Heavy Sections or Fatigue (see 10.13.1)

5_ Note varie

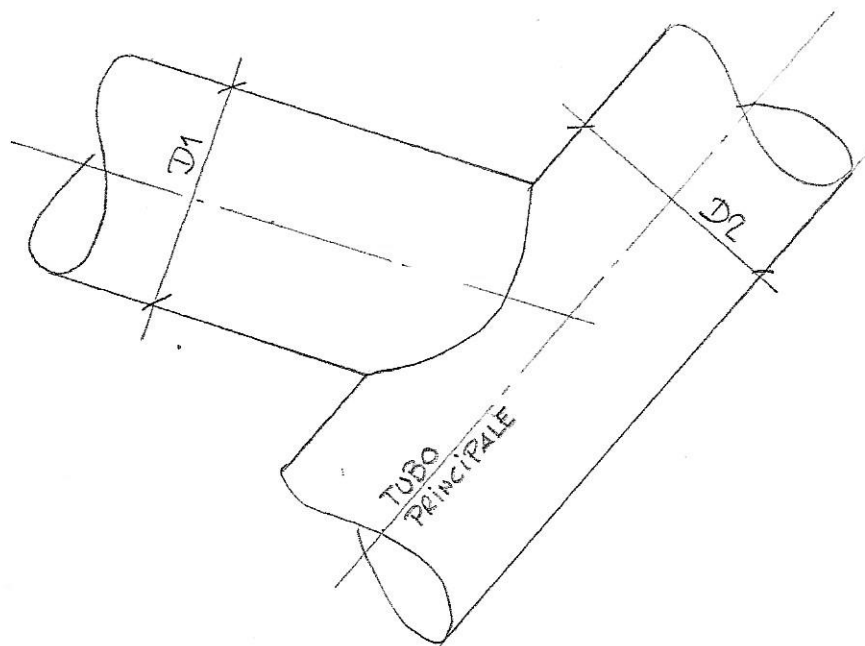
a) Grado di lavorazione superficiale dopo il taglio

Sezionando un tubo con un taglio inclinato si possono notare due differenti gradi di lavorazione superficiale. La peggiore la possiamo riscontrare sul taglio con angolo negativo.

Minore e' l'angolo di taglio peggiore e' la superficie.



b) Nella programmazione di innesti con lo stesso diametro e' buona norma aumentare dell' 1-2% il diametro del tubo principale.



$$\text{SE } D1 = D2$$

$$D2 = D2 + 1-2\% D2$$